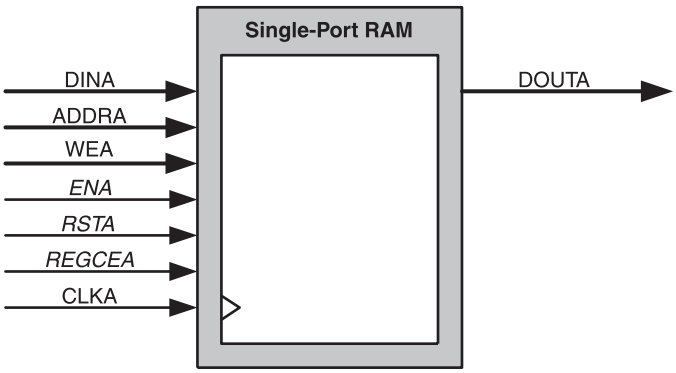



*Figure 3-3:* **Single-port RAM**

The Simple Dual-port RAM provides two ports, A and B, as illustrated in Figure 3-4. Write access to the memory is allowed through port A, and Read access is allowed through port B.

**IMPORTANT:** *For 7 series family architectures, though the diagram below shows port A as write port and port B as Read Port, in the FPGA the BRAM primitives perform Read access* through *port A and Write access* through *port B.*

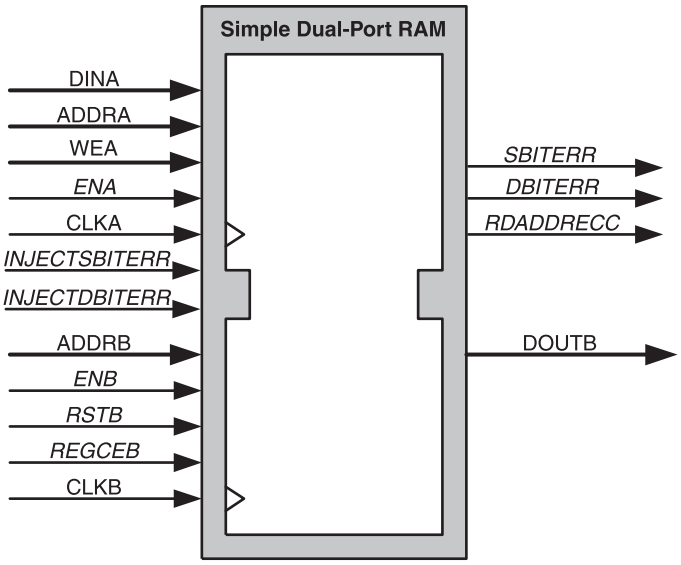



*Figure 3-4:* **Simple Dual-port RAM**

The True Dual-port RAM provides two ports, A and B, as illustrated in Figure 3-5. Read and Write accesses to the memory are allowed on either port.

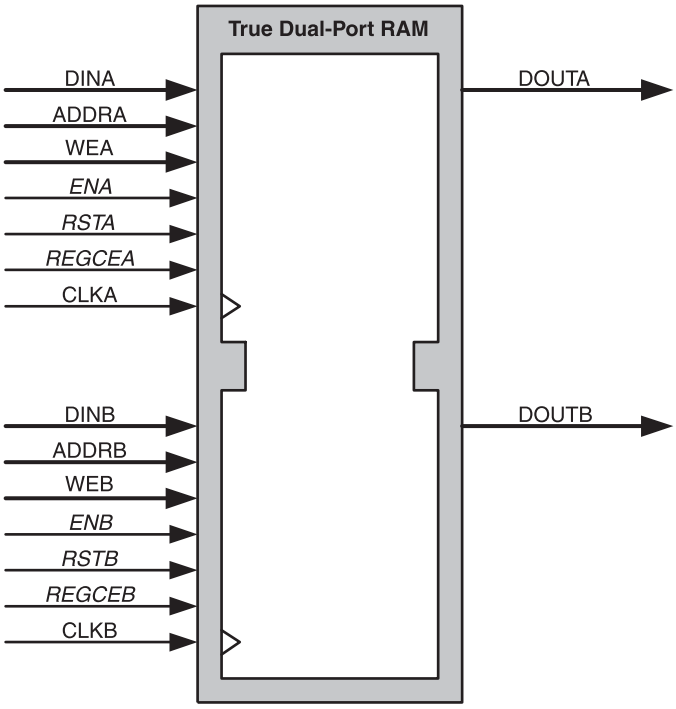


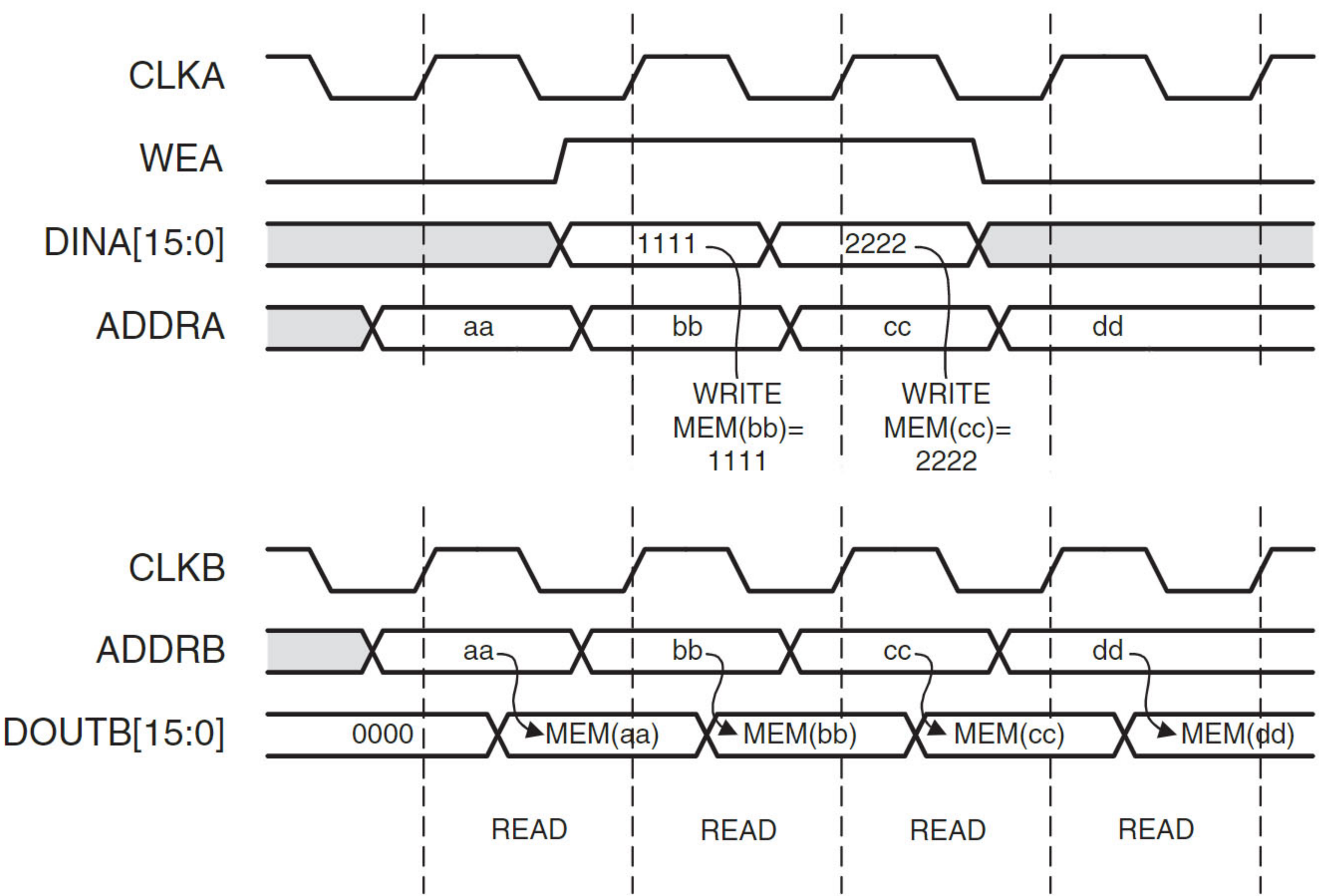

*Figure 3-5:* **True Dual-port RAM**

## Selectable Memory Algorithm

The Block Memory Generator core arranges block RAM primitives according to one of three algorithms: the minimum area algorithm, the low power algorithm and the fixed primitive algorithm.

### *Minimum Area Algorithm*

The minimum area algorithm provides a highly optimized solution, resulting in a minimum number of block RAM primitives used, while reducing output multiplexing. Figure 3-6 shows two examples of memories built using the minimum area algorithm.


CLKA
WEA
DINA[15:0]
ADDRA
1111
2222
aa
bb
cc
dd
WRITE MEM(bb)= 1111
WRITE MEM(cc)= 2222
CLKB
ADDRB
DOUTB[15:0]
0000
MEM(aa)
MEM(bb)
MEM(cc)
MEM(dd)
READ
READ
READ
READ